Fiscal Year:	FY 2026	**Task Last Updated:**	FY 08/04/2025
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
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Key Personnel Changes/Previous PI:			

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Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
	The health risks associated with spaceflight-induced ocular structural and functional damage has become a recent concern for NASA. Our research goal of the present study was to characterize the effects of spaceflight and reentry to 1 g on the structure and integrity of the retina and blood-retinal barrier (BRB) in the eye. To investigate possible mechanisms, changes in protein expression profiles were examined in mouse ocular tissue after spaceflight. Ten-week-old male C57BL/6 mice were launched to the International Space Station (ISS) on SpaceX CRS-12 (SpaceX-12) at the NASA Kennedy Space Center (KSC) in August 2017. After a 35-day mission, mice were returned to Earth alive. Within 38 +/- 4 hours of splashdown, ocular tissues were collected for analysis. Ground control (GC) and vivarium control mice were maintained on Earth in flight hardware or normal vivarium cages, respectively. Repeated intraocular pressure (IOP) measurements were performed before the flight launch and re-measured before the mice were sacrificed after splashdown. IOP was significantly lower in post-flight measurements compared to that of pre-flight. The flight group had significant apoptosis in the retina and retinal vascular endothelial cells compared to the control groups. Immunohistochemical analysis of the retina revealed changes in biomarker expressions related to BRB integrity. Proteomic analysis showed that many key proteins and pathways responsible for cell death, cell cycle, immune response, mitochondrial function, and metabolic stress were significantly altered in the flight mice compared to ground control animals. These data indicate a complex cellular response that may alter retina structure and BRB integrity following long-term spaceflight. Extended spaceflight has been shown to adversely affect astronaut visual acuity. The purpose of this study was to determine whether spaceflight alters gene expression profiles and induces oxidative damage in the retina. Ten-week-old adult C57BL/6 male mice were flown aboard the ISS for 35 days and returned to Earth alive. Ground control mice were maintained on Earth under identical environmental conditions. Within 38 (+/-4) hours after splashdown, mice ocular tissues were collected for analysis. RNA sequencing detected 600 differentially expressed genes (DEGs) in murine spaceflight retinas, which were enriched for genes related to visual perception, the phototransduction pathway, and numerous retina and photoreceptor phenotype categories. Twelve DEGs were associated with retinitis pigmentosa, characterized by dystrophy of the photoreceptor layer rods and cones. Differentially expressed transcription factors indicated changes in chromatin structure, offering clues to the observed phenotypic changes. Immunofluorescence assays showed degradation of cone photoreceptors and increased retinal oxidative stress. Total retinal, retinal pigment epithelium, and choroid layer thickness were significantly lower after spaceflight. These results indicate that retinal performance may decrease over extended periods of spaceflight and cause visual impairment. Astronauts returning from spaceflight have experienced eye problems, which may decrease retinal performance and lead to long-term effects on visual acuity. This study leverages the collected data from spaceflown murine retinas that were treated with redox-active metalloporphyrin (BuOE) to mitigate spaceflight-induced changes and respective ground controls. 10-week-old adult C57BL/6 male mice were flown on SpaceX-24 (RR-18) to the ISS national lab, kept in low Earth orbit for 35 days, and returned to Earth alive. Our multi-omics analysis of RNA-sequencing (RNA-Seq) and reduced representation bisulfite sequencing (RRBS) data generated from subsequent murine retina tissues uncovered genes, pathways, and epigenetic modifications consistent with the therapeutic potential of BuOE. From RNA-Seq

Task Progress:	analysis of spaceflown murine samples, the treatment group shows differentially expressed genes relative to saline controls that reached significance, and genes that are related to protection against cell oxidative damage and cellular response to organonitrogen compounds. Ranked fold-changes from the same contrast were used for gene set enrichment analysis, which showed biological processes reaching significance, including glutathione metabolic processes and cellular response to xenobiotic stimulus. The findings from this investigation have the potential to provide valuable insights into the molecular mechanisms underlying conditions like Spaceflight Associated Neuro-ocular Syndrome (SANS) and assess the effectiveness of BuOE as a countermeasure for astronauts experiencing neuro-ophthalmic abnormalities, which can lead to long-term effects on visual acuity. The physical demands of deep-space pose a health risk to the central nervous system that has long been a concern when sending humans to space. While little is known about how spaceflight affects transcription spatially in the brain, understanding this has the potential to guide strategies to mitigate the effects of spaceflight on the brain. Therefore, we performed Digital Spatial Profiling (DSP) of mouse brains subjected to either spaceflight, or grounded controls. Four brain regions were selected for DSP: Cortex, Frontal Cortex, Corunu Ammonis 1 (CA), and Dentate Gyrus (DG). Because spaceflight is associated with increased oxidative stress, treatment with antioxidants has emerged as a potential strategy to attenuate the effects of spaceflight. We therefore treated a subset of the spaceflight and control mice with the superoxide dismutase mimic, MnTnBuOE-2-PyP 5+ (BuOE), and performed DSP on them as well. Our analysis revealed hundreds of differentially expressed genes (DEGs) due to spaceflight in all four regions of frontal cortex, cortex, CA1, and DG, with CA having the highest number of DEGs. Each brain region showed a distinct transcriptomic response, while some DEGs were common to two or more regions. A pathway analysis revealed distinct pathways sensitive to oxidative stress, as well as metabolic pathways, to be enriched in each of the brain regions. These represent the first spatial gene expression profiles taken of rodents in space, and may help improve our understanding of brain region-specific susceptibility to spaceflight conditions. Treatment with BuOE reduced the transcriptomic effects of spaceflight at a large number of DEGs, suggesting that BuOE may attenuate oxidative stress-induced brain damage. The goal of the present study was to characterize the effects of spaceflight and reentry to 1g on the acute and late intraocular pressure (IOP) changes and function of the retina. To determine the extent to which an antioxidant protects against spaceflight environment-induced changes in the retina, the mice were administered weekly superoxide dismutase mimic, MnTnBuOE-2-PyP 5+ (BuOE) or saline treatments, before launch and during their stay aboard the ISS. Forty ten-week-old adult C57BL/6 male mice were flown aboard the ISS on SpaceX-24 over 35 days and returned to Earth alive. Ground control mice were maintained on Earth under identical environmental conditions. Before launch, IOP was measured using a handheld tonometer, and retinal function was evaluated using electroretinogram (ERG). ERG data was recorded when the mouse eye was under dark-adapted conditions in response to a weak, diffuse, full-field flash of light. Within 20 hours after splashdown, IOP and ERG assessments were repeated before sacrifice. Subsets of mice will be measured after 6 months of the spaceflight for the re-adaptation study. There were significant increases in body weight for both habitat control and vivarium control groups post-flight compared to pre-flight measurements ($p<0.05$). However, the body weights were similar among flight groups before launch and after splashdown. The IOP measurements were similar between pre-and post-flight groups. There were no significant differences between BuOE-treated mice and saline controls. To assess the retinal function, we measured the amplitude of the scotopic ERG b-wave, which is primarily generated by ON-type bipolar cells. Bipolar cells are the main neurons of the retina that carry signals from photoreceptors to ganglion cells. These data indicate, for the first time, spaceflight conditions may induce bipolar cell damage and dysfunction. It has been proposed that neuroinflammatory response plays an important role in the neurovascular remodeling in the brain after stress. The goal of the present study was to characterize changes in the gene expression profiles associated with neuroinflammation, neuronal function, metabolism, and stress in mouse brain tissue. Ten-week-old male C57BL/6 mice were launched to the International Space Station (ISS) on SpaceX-12 for a 35-day mission. Within 38 ± 4 h of splashdown, mice were returned to Earth alive. Brain tissues were collected for analysis. A novel, digital color-coded barcode counting technology was used to evaluate gene expression profiles in the spaceflight mouse brain. A set of 54 differently expressed genes significantly segregates the habitat ground control (GC) group from the flight (FLT) group. Many pathways associated with cellular stress, inflammation, apoptosis, and metabolism were significantly altered by flight conditions. A decrease in the expression of genes important for oligodendrocyte differentiation and myelin sheath maintenance was observed. Moreover, mRNA expression of many genes related to anti-viral signaling, reactive oxygen species (ROS) generation, and bacterial immune response was significantly downregulated. Here, we report that significantly altered immune reactions may be closely associated with spaceflight-induced stress responses and have an impact on neuronal function.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Articles in Peer-reviewed Journals	Roy U, Hadad R, Rodriguez AA, Saju A, Roy D, Gil M, Keane RW, Scott RT, Mao XW, de Rivero Vaccari JP. "Effects of spaceflight on inflammasome activation in the brain of mice." Cells. 2025 Mar 12;14(6):417. https://doi.org/10.3390/cells14060417 ; PMID: 40136666; PMCID: PMC11941215 , Mar-2025
Articles in Peer-reviewed Journals	Patel CM, Wiele SV, Kim L, Payne E, Bruno-Garcia M, Devorak A, Kaganov DE, Lau A, Guthold M, Delp MD, Crapo J, Mao XW, Willey JS. "Treatment with a superoxide dismutase mimetic for joint preservation during 35 and 75 days in orbit aboard the International Space Station, and after 120 days recovery on Earth." Life Sci Space Res (Amst). 2025 Feb:44:67-78. https://doi.org/10.1016/j.lssr.2024.10.009 ; PMID: 39864914 , Feb-2025
Articles in Peer-reviewed Journals	Kremsky I, Ali S, Stanbouly S, Holley J, Justinen S, Pecaut M, Crapo J, Mao X. "Spaceflight-induced gene expression profiles in the mouse brain are attenuated by treatment with the antioxidant BuOE." Int J Mol Sci. 2023 Sep 1;24(17):13569. https://doi.org/10.3390/ijms241713569 ; PubMed PMID: 37686374; PubMed Central PMCID: PMC10487739 , Sep-2023

Articles in Peer-reviewed Journals	Casaletto JA, Scott RT, Rathod A, Jain A, Chandar A, Adapala A, Prajapati A, Nautiyal A, Jayaraman A, Boddu A, Kelam A, Jain A, Pham B, Shastry D, Narayanan D, Kosaraju E, Paley E, Uribe FP, Shahid I, Ye I, Wu J, Lin J, Srinivas K, Paolieri Della Monica M, Hitt M, Lin M, Volkan M, Kharya M, Kaul M, Jaffer MA, Ali M, Chang NZ, Ashri N, Couderc NB, Paladugu P, Sood R, Hiremath R, Pathak R, Dogra S, Srinivas S, Samaddar S, Gopinath S, Sawant S, Cai S, Pala V, Nair V, Shi Z, Anand Narayanan S, Thomas DM, Lewkowicz A, Waisberg E, Ong J, Gebre S, Galazka JM, Vaishampayan PA, Sanders LM, Mao XW. "Machine learning ensemble reveals distinct molecular pathways of retinal damage in spaceflown mice." npj Microgravity. 2026 Jul 3. Early access article. https://doi.org/10.1038/s41526-026-00625-w ; PMID: 42399270 , Jul-2026